FINAL PROJECT PROPOSAL

Explore Ames

PREPARED FOR

COM S 3190 - Construction of User Interfaces
Iowa State University Computer Science Department

PREPARED BY

Anish Reddy Nukala
Prajwal Reddy Chenreddy

Contents

1 Introduction

2 Purpose of the Proposal

3 Goals & Objectives

4 Project Description

5 Exercising Project Options

6 Feature Ownership & Responsibility

7 Resources and Tools

8 Data Sources and Management

9 AI Features

10 User Experience Views

11 Final Comments

1 Introduction

We, Anish Reddy Nukala and Prajwal Reddy Chenreddy, are members of team NM_05. Both of us are sophomores majoring in computer science. Through COM S 3190, we are developing important web development skills such as HTML, CSS, and JavaScript. From the in-class activities, we have gained experience in building a product catalog website. We believe that by combining our ideas and skills, we can create an even better website.

2 Purpose of the Proposal

We aim to develop a website that serves as both an informative and engaging hub for those interested in exploring Ames. Through our website Explore Ames, we want to showcase the city's culture, history, and attractions. We believe Ames is a place with much to offer and can leave a lasting impact on its visitors. Our goal is to design an interactive website that allows users to discover and experience Ames in a meaningful way

3 Goals & Objectives

Goals:

- Build a full-stack, responsive web application using React and Node.js.
- Improve team skills in frontend/backend integration.
- Design an intuitive and welcoming UI/UX.

Objectives:

- **Week 1 (Nov 10 – Nov 16):** Set up GitLab repo, initialize project structure.
- **Week 2 (Nov 17 – Nov 23):** Design more wireframes and UI mockups, Implement routing and homepage UI.
- **Week 3 (Nov 24 – Nov 30):** Build backend APIs.
- **Week 4 (Dec 1 – Dec 7):** Implement core features (CRUD, navigation).
- **Week 5 (Dec 8 – Dec 14):** Perform user testing, polish UI, and push to repo.

4 Project

With our project Explore Ames, we intend to create a website dedicated to showcasing the city of Ames. We believe Ames is a vibrant place with a unique culture, history, and community spirit. Our aim is to present the best of Ames from its past to its present while also serving as a practical guide for visitors and residents to explore what the city has to offer.

- Frontend: Built in React with React Router for navigation and CSS for styling.
- Backend: Developed using Node.js, mongoDB and Express to handle API routes and business logic.
- Database: MongoDB Atlas for cloud-based storage of users, reviews, and favorites.
- APIs: OMDb API for explore data.

Planned Pages (6 Total):

1. **Home:** Quick navigation to all sections
2. **Diners:** Local food spots with ratings & details
3. **Explore Us:** Map and famous places
4. **Shop:** Ames-themed merch (preview only)
5. **Authors:** About the student team
6. **FAQ:** Common questions about the project

5 Exercising Project Options

Extending Midterm Project.

This project will be a extended implementation of our Midterm Project created from scratch again using React, Node.js, and MongoDB. We plan to:

Implementation Plan:

- Start with UI design in Excalidraw.
- Create frontend components and pages in React.
- Build backend API with Express and connect to MongoDB.
- Use RESTful API communication between frontend and backend.

6 Feature Ownership & Responsibility

Feature	Description	Assigned Developer	Tech Involvement
Login & Signup	User authentication and session management	Anish N	Frontend + Backend
Explore Us	Interactive map with location info	Prajwal C	Frontend + Backend
Shop	Ames-themed merch	Anish N	Frontend + Backend
Diners	Dining location and reviews	Prajwal C	Frontend + Backend
Authors	About the student team	Anish N	Frontend + Backend
FAQ's	Common questions about the project	Prajwal C	Frontend + Backend

7 Resources and Tools

Tech Stack:

- React, React Router, Axios
- Node.js, Express
- MongoDB
- GitLab for version control
- Excalidraw for design

Time Commitment:

- Each team member commits ~8 hours/week
- Weekly stand-up meetings and task distribution via GitLab issues

8 Data Sources and Management

The Explore Ames website uses structured and dynamic data to manage content effectively across different sections. These data sources are organized using JSON format and stored locally within the project directory for fast and flexible content rendering. Key data categories include:

- **Places**: High-level navigation cards that represent different sections of the website such as Diners, Explore Us, Shop, and About. Each entry contains a title, an image path, short chip tags (emoji-labeled categories), a description, a button label, and a link to the corresponding page.
- **Diners**: A detailed list of restaurants in Ames including key attributes like cuisine type, specialty items, business hours, full address, rating, and an image reference. This data helps dynamically generate restaurant cards for the Diners page.
- **Explore**: Featured places of interest in Ames such as parks, landmarks, and scenic locations. Each place includes a name, a brief description, and an image, allowing users to visually explore local attractions

Sample JSON Structures:

1. places.json – Used to populate the homepage with navigation cards:

```
{
  "title": "Diners",
  "image": "assets/images/diners.jpg",
  "chips": ["🍳 Breakfast", "🍔 Casual"],
  "description": "From hearty breakfasts to late-night bites, enjoy Ames' best local food spots.",
  "link": "diners.html",
  "buttonText": "Explore Diners"
}
```

2. diners.json – Used to build restaurant cards dynamically on the Diners page:

```
{
  "name": "The Café",
  "cuisine": "American",
  "specialty": "Brunch & Coffee",
  "timings": "8 AM – 2 PM",
  "address": "2616 Northridge Pkwy, Ames, IA",
  "rating": "⭐⭐⭐⭐⭐",
  "image": "assets/images/diner1.png"
}
```

Data Collections:

- **Places Collection**:

  Stores high-level cards for categories like Diners, Explore Us, Shop, and About — including title, image, description, category tags, and navigation links.

- **Restaurants Collection**:

  Stores detailed information for each diner such as name, cuisine type, specialty, operating hours, address, rating, and images.

- **Attractions Collection**:

  Contains information about local attractions (e.g., parks, landmarks, trails) with names, descriptions, and visual assets.

- **Users Collection (Future)**:

  Will store user accounts, preferences, submitted reviews, and saved favorites.

- **Reviews Collection (Future)**:

  Will link user feedback and ratings to specific restaurants or places.

Data Origin and Flow:

- **Initial content** (e.g., diners, attractions, shops) is created and managed manually via structured JSON files and will later be migrated to MongoDB for persistent storage.

- **User data** such as submitted reviews or feedback (planned) will be collected via interactive forms in the frontend and processed through backend APIs.

- All **CRUD operations** (Create, Read, Update, Delete) will be managed securely using the MongoClient library within a **Node.js/Express** backend.

Data Management Workflow:

1. A user interacts with the Explore Ames frontend — for example, browsing diners or submitting a review (in future versions).
2. The React-based interface sends the user request via Axios to the appropriate Express.js API route.
3. The backend validates and processes the request, then performs the appropriate MongoDB operation (insert, find, update, delete) using the Mongo Client library.
4. The backend returns a JSON response, which is dynamically rendered on the frontend without requiring a full page reload.
5. Admin-level users may also interact with a dashboard for content updates — to manage new entries, update timing changes, or moderate reviews.

9 AI Features

To make *Explore Ames* smarter and more helpful, we plan to add two simple AI features:

- **Place Suggestions**: Based on what users click on (like diners or parks), the site can recommend similar places they might enjoy.
- **Short Descriptions**: Use AI to turn long texts into short summaries, so users can read quickly without missing key info.

These small features will make the website feel more personalized, while still being realistic to build during the project.

10 User Experience Views

Example Wireframe for Homepage:

- A welcoming interface with clear navigation to major sections such as History, Gallery, Shop, and About.
- Engaging visual elements like sliding images that highlight the beauty, culture, and attractions of Ames.
- Interactive categories such as Diners, Explore Us, Shop Us, and About, each with short descriptions and visuals to guide users.
- A clean and user-friendly layout that makes the site easy to explore and encourages visitors to learn more about Ames.

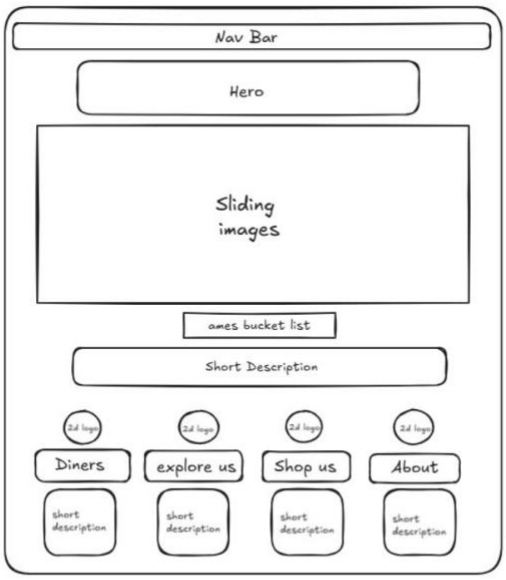

Figure 1: Home Page Wireframe - Designed on Excalidraw

Example Wireframe for Homepage:

- A welcoming interface with clear navigation to major sections such as History, Gallery, Shop, and About.
- Engaging visual elements like sliding images that highlight the beauty, culture, and attractions of Ames.
- Interactive categories such as Diners, Explore Us, Shop Us, and About, each with short descriptions and visuals to guide users.
- A clean and user-friendly layout that makes the site easy to explore and encourages visitors to learn more about Ames.

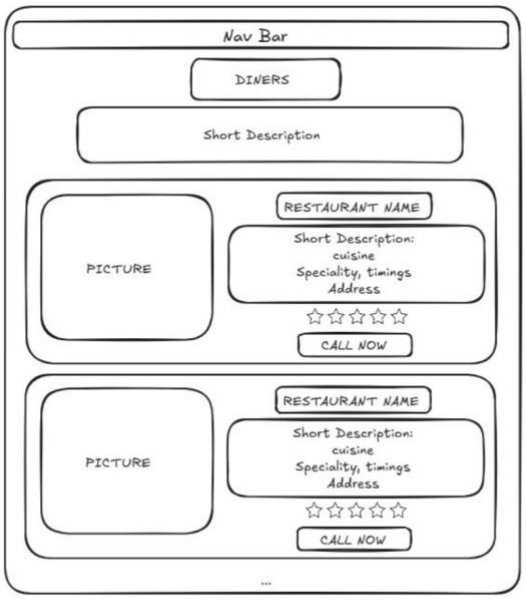

Figure 2: Diners Page Wireframe - Designed on Excalidraw

11 Final Comments

This is our proposal for Explore Ames. We believe this project will not only strengthen our web development skills but also serve as a valuable resource for both residents and visitors of Ames. Thank you for considering our proposal. We welcome any feedback or suggestions to improve our work. You can reach out to us at: arnukala@iastate.edu or preddy@iastate.edu